1)

$p = 0{,}2$

a) Geometrisk : $f_Z(z) = p(1-p)^{x-1}$, $F_Z(z) = 1-(1-p)^x = P(Z \le z)$

b)

$$P(Z \ge 3) = 1 - P(Z \le 2) \quad | \text{diskret}$$

$$= (0{,}8)^2 = 0{,}64$$

c) Trenger 8, W = #pers må ringe før $8x(Z=1)$

W = neg. binomisk

d) $E[W] = \frac{k}{p}$, $f_W(w) = \binom{w-1}{k-1} p^x (1-p)^{w-k}$

$k = 8$, $p = 0{,}2$

$E[W] = \frac{8}{0{,}2} = \underline{40}$

e) $SD[W] = \sqrt{Var[W]}$

$Var[W] = \frac{(1-p)k}{p^2} = \frac{0{,}8 \cdot 8}{0{,}04} = \frac{6{,}4 \cdot 100}{4} = \underline{160}$

$SD[W] = \sqrt{160}$

f) ringer 20 pers pr dag

X = # deltakere (til spørreund.) som ansatt gråter i skaffe

Binomisk

$f_X(x) = \binom{n}{x} p^x (1-p)^{n-x}$, n = # samtaler $\quad n = 20 = 1$ dag

p = % for deltaker

x = # deltakere

$W = \frac{\#\text{ ringt}}{\#\text{ deltakere}}$, $W > 20 \mid k = 8$

$X = \frac{x \#\text{ deltakere}}{20 \text{ ringt}}$? $P(X < 8) \quad \mid n = 20$

$n = 20 = 1$ dag

$P(n > 20 \mid x = 8)$ rart $\rightarrow = P(X \leq 7) \quad \mid p = 0,2$

$\rightarrow$ Tabell $= 0,968$

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2) $\lambda = 5,5$ h pr time

$X \sim \text{poisson}(\lambda)$, # h i løpet av 1 time

a) $P(X = 4) \quad (\lambda t = \lambda = 5,5)$

$= P(X \leq 4) - P(X \leq 3)$

$= 0,3575 - 0,2017$

$= 0,1558$

c) Failrate = 1 pr min $= \lambda$

Y = # fail i løpet av 4 min

$E[Y] = \lambda t = 1 \cdot 4 = \underline{4}$

b) $P(Y = 3) = P(Y \leq 3) - P(Y \leq 2)$

$= 0{,}4335 - 0{,}2381$

$= \underline{0{,}1954}$

c) $P(Y > 3) = 1 - P(Y \leq 3)$

$= \underline{\underline{0{,}5665}}$

3)

8 ml virkestoff + 100 ml vann

fortynnes 1:100 ×12

Y = # molekyler av virkestoff i produkt

n = # molekyler $= 8 \cdot N_A$

p = % for partikkel i overføre 12 fortynninger $= \left(\frac{\frac{1}{100}}{100}\right) = \left(\frac{1}{100}\right)^{12} = \underline{\underline{1 \cdot 10^{-24}}}$ (×12)

$E[Y] = np = 8 \cdot 6{,}022e23 \cdot 1e{-}24 = 4{,}8176$

f) $P(Y > 0) = 1 - P(Y \leq 0)$ $\quad | P(Y = 0)$

4) Z är normalfördelad stok. var.

$$P(Z \leq 1,7) = 0,9554$$

$$P(0,2 \leq Z \leq 1,7) = P(Z \leq 1,7) - P(Z \leq 0,2) = 0,9554 - 0,5793$$

$$= 0,3761$$

$\mu = 1,3$, $\sigma = 2,3$

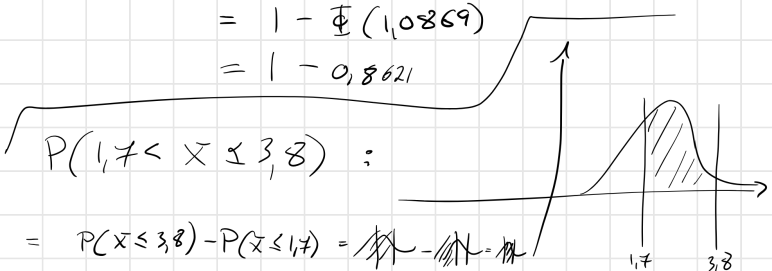
$$P(X > 3,8) = 1 - P(X \leq 3,8)$$

$$= 1 - \Phi\left(\frac{3,8 - 1,3}{2,3}\right)$$

$$= 1 - \Phi(1,0869)$$

$$= 1 - 0,8621$$

$P(1,7 < X \leq 3,8)$:



$$= P(X \leq 3,8) - P(X \leq 1,7) = \text{[illegible]}$$

$$= 0,8621 - \Phi\left(\frac{1,7 - 1,3}{2,3}\right)$$

$$= 0,8621 - \Phi(0,1739)$$

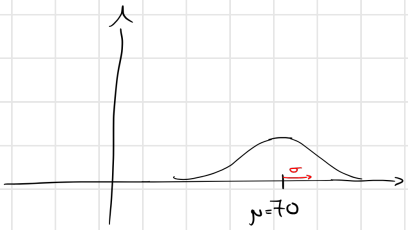
$$= 0,8621 - 0,5675$$

$$= 0,2946$$

5] $\mu = 70$, $\sigma = 6$

$\bar{x}$ = fart

$P(\bar{x} > x_\alpha) = \alpha$



pusti stopper kun 20% raskeste = top 20%

$x_\alpha = (1-\alpha)$-kvantil , $\alpha = 0{,}2$?

$x_\alpha = 0{,}842$

$x_\alpha = \frac{z-\mu}{\sigma} \Rightarrow x_\alpha \sigma + \mu = z$

$(\sigma \cdot x_\alpha) + \mu = 75{,}052$

$\Phi(x_\alpha) = F_X(z)$ for standard normalfordeling ($\Phi(z)$ sies ui)

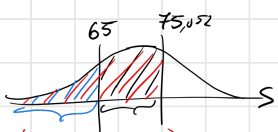
$\varphi(x_\alpha) = f_X(z)$

b) $P(\bar{x} > 80 \mid \bar{x} > 75{,}052) = \frac{P(\bar{x} > 80 \cap \bar{x} > 75)}{P(\bar{x} > 75)} = \frac{P(\bar{x} > 80)}{P(\bar{x} > 75)}$

$= \frac{1 - P(\bar{x} \le 80)}{1 - P(\bar{x} \le 75)} = \frac{1 - \Phi\left(\frac{80-70}{6}\right)}{1 - \Phi\left(\frac{75-70}{6}\right)} = \frac{1 - \Phi(1{,}66)}{1 - \Phi(0{,}83)} \approx \Phi(1{,}7)$

$= \frac{1 - 0{,}9525}{1 - 0{,}7967} = 0{,}2336$

c) $P(\bar{x} > 65 \mid \bar{x} < 75{,}052)$



$\bar{z} = z - z$

$P(A \cap B) = P(\bar{x} \le 75{,}052) - P(\bar{x} \le 65)$

$(1-\alpha) - \Phi\left(-\frac{5}{6}\right)$

$= 0{,}800 - 0{,}2033 = 0{,}5962$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{0,5962}{0,8} = 0,7459$$

6) $\mu = 54 \quad \sigma = 2$ | poängsum

35% = kvinnor får +1 poäng

poänggräns = 56

a) $P(X_k > 56) = P(X > 55) = 1 - P(X \leq 55)$

$= 1 - \Phi\left(\frac{1}{2}\right) = 1 - 0,6915$

$= 0,3085$

b) $P(X > 56) = 1 - P(X \leq 56) = 1 - \Phi(1) = 1 - 0,8413$

$= 0,1587$

c) $P(\text{kvinna} \mid Z > 56)$, $Z = \begin{cases} x+1, & \text{kvinna} \\ x, & \text{man} \end{cases}$

$$= \frac{P(\text{kvinna}) \times P(x > 55 \mid \text{kvinna})}{P(Z > 56)} = \frac{P(\text{kvinna}) \times P(X > 55)}{P(X > 56 \mid \text{kvinna})\,P(k) + P(X > 56 \mid m)\,P(m)} = \frac{0,35 \cdot 0,3085}{[0,3085 \times 0,35 + 0,1587 \cdot 0,65]} = 0,5114$$

d)

$P(X > 61 \mid k) = P(X > 60) = 1 - \Phi(3) = 0,0013$

$P(X > 61) = 1 - \Phi\left(\frac{61-54}{2}\right) = 0,0002$

$$\frac{0,35 \cdot 0,0013}{0,35 \times 0,0013 + 0,65 \cdot 0,0002} = 0,77$$